

DOCTOR APPOINTMENT BOOKING SYSTEM USING ML

A PROJECT REPORT

Submitted by

THANGA KAVIYA G (8115U24AM057)

in partial fulfilment for the award of the course

AMB 1221-MACHINE LEARNING FUNDAMENTALS

IN

DEPARTMENT OF

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)



**K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY**



**ANNA UNIVERSITY
CHENNAI 600 025**

MAY 2026

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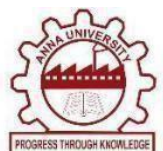
Department of Artificial Intelligence and Data Science

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BONAFIDE CERTIFICATE

Certified that this project report titled **“DOCTOR APPOINTMENT BOOKING SYSTEM USING ML”** is the Bonafide work of **THANGA KAVIYA G (8115U24AM057)** who carried out the work under my supervision.

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DECLARATION BY THE CANDIDATE

I declare that to the best of my knowledge the work reported here in has been composed solely by myself and that it has not been in whole or in part in any previous application for a degree.

Submitted for the project Viva-Voice held at K. Ramakrishnan College of Engineering on _____

SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

I thank the almighty GOD, without whom it would not have been possible for me to complete my project.

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K. RAMAKRISHNAN COLLEGE OF ENGINEERING

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ANNA UNIVERSITY, CHENNAI



DEPARTMENT OF CSE (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

VISION

To become a renowned hub for AIML technologies to producing highly talented globally recognizable technocrats to meet industrial needs and societal expectation.

MISSION

Mission of the Department

- M1** To impart advanced education in AI and Machine Learning, built upon a foundation in Computer Science and Engineering.
- M2** To foster Experiential learning equips students with engineering skills to tackle real-world problems.
- M3** To promote collaborative innovation in AI, machine learning, and related research and development with industries.
- M4** To provide an enjoyable environment for pursuing excellence while upholding strong personal and professional values and ethics.

PROGRAM EDUCATIONAL OBJECTIVES (PEO's)

- PEO1** Excel in technical abilities to build intelligent systems in the fields of AI & ML in order to find new opportunities.
- PEO2** Embrace new technology to solve real-world problems, whether alone or as a team, while prioritizing ethics and societal benefits.
- PEO3** Accept lifelong learning to expand future opportunities in research and product development.

PROGRAM SPECIFIC OUTCOMES (PSO's)

- PSO1** Expertise in tailoring ML algorithms and models to excel in designated applications and fields.
- PSO2** Ability to conduct research, contributing to machine learning advancements and innovations that tackle emerging societal challenges.

PROGRAM OUTCOMES (POs)

Engineering students will be able to:

PO1 Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2 Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3 Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4 Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5 Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.

PO6 The Engineer and Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7 Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO8 Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings

PO9 Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10 Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11 Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

ABSTRACT

The Doctor Appointment Booking System using ML is designed to provide an intelligent and efficient solution for managing medical appointments. It helps to overcome problems like long waiting times and manual booking errors. The system uses machine learning techniques to improve scheduling and decision-making. It analyzes important factors such as patient history, doctor availability, specialization, and appointment priority. Based on this data, it predicts suitable time slots for patients. It also recommends the most appropriate doctors according to user requirements. The system is developed as a web-based application using Python and the Flask framework. It provides an easy-to-use and user-friendly interface for patients and administrators. The application includes modules like user registration, login, doctor search, and appointment booking. It also supports schedule management and notification features. Users can quickly book appointments and receive instant confirmation. The system helps in reducing overcrowding and improves hospital workflow. It delivers fast and accurate real-time results with low computational cost. Overall, it shows the effective use of machine learning in healthcare with strong potential for future enhancements.

ABSTRACT WITH POs AND PSOs MAPPING

ABSTRACT	POs MAPPED	PSOs MAPPED
The Doctor Appointment Booking System using AI is developed to simplify and automate the process of scheduling medical consultations using machine learning techniques. It analyzes factors such as patient history, doctor availability, specialization, time slots, and urgency of symptoms to provide smart appointment recommendations. The system uses a Machine Learning model (such as classification or regression) to predict suitable doctors and optimal booking times. It is implemented as a web-based application using Python, enabling users to book appointments easily, reduce waiting time, and improve healthcare service efficiency.	PO1-3 PO2-3 PO3-3 PO4-3 PO5-3 PO6-3 PO7-3 PO8-3 PO9-3 PO10-3 PO11-3	PSO1-3 PSO1-3

Note: 1-Low, 2-Medium, 3-High

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LIST OF ABBREVIATIONS

S.NO.	ACRONYM	ABBREVIATION
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	DBMS	Database Management System
4	GUI	Graphical User Interface
5	ML	Machine Learning
6	SQL	Structured Query Language
7	UI	User Interface

CHAPTER 1

INTRODUCTION

1.1 Objective

The objective of the Doctor Appointment Booking System using ML is to develop an intelligent and efficient platform for managing medical appointments. It aims to reduce waiting time and improve patient convenience through automated scheduling. The system focuses on analyzing patient data, doctor availability, and specialization to optimize appointment allocation. It uses machine learning techniques to predict suitable time slots and recommend appropriate doctors. Another objective is to minimize manual errors and improve accuracy in booking processes. The system also aims to enhance communication between patients and healthcare providers. It provides a user-friendly interface for easy access and smooth interaction.

The project ensures secure user authentication and data management. It helps hospitals manage appointments more effectively and reduce overcrowding. The system delivers fast and real-time responses with minimal computational cost. It supports better decision-making for both patients and administrators. It also aims to improve overall healthcare service quality and efficiency. The project demonstrates the practical application of machine learning in healthcare systems. Overall, it provides a smart and scalable solution for modern appointment management.

1.2 Overview

The Doctor Appointment Booking System using ML is a smart web-based application designed to simplify and automate the process of scheduling medical appointments. It uses machine learning techniques to analyze patient details, doctor availability, specialization, and previous booking patterns to provide optimized appointment slots. The system allows users to register, log in, search for doctors, and book appointments بسهولة through an easy-to-use interface. It also recommends suitable doctors.

This is where the system becomes truly smart. A no-show prediction model analyzes historical patterns — such as a patient's booking history, appointment type, day of the week, and weather — to estimate the likelihood of a missed appointment, allowing the system to strategically overbook low-risk slots. A doctor-matching engine uses collaborative filtering and content-based recommendation to suggest the most suitable doctor based on specialty, patient ratings, location, and past interactions. A wait-time prediction model uses queue data and doctor pace metrics to give patients accurate estimated wait times. An NLP-based symptom triage module reads free-text symptom descriptions entered by patients and assigns an urgency score, helping route critical cases to same-day slots automatically.

The application is developed using Python and the Flask framework, ensuring efficient performance and accessibility. It includes key modules such as user authentication, doctor selection, appointment booking, schedule management, and notification system. The ML component enhances the system by reducing waiting time, avoiding scheduling conflicts, and improving overall efficiency. The system provides real-time responses and ensures smooth interaction between patients and healthcare providers. Overall, it offers a reliable, fast, and user-friendly solution for modern healthcare appointment management.

1.3 Purpose and Importance

The purpose of the Doctor Appointment Booking System using ML is to provide an efficient and intelligent solution for scheduling medical appointments. It aims to simplify the booking process by allowing patients to easily find doctors and reserve suitable time slots. The system reduces manual work and minimizes errors in appointment management. By using machine learning techniques, it helps in predicting the best available time slots and recommending appropriate doctors based on patient needs.

The importance of this system lies in improving healthcare service efficiency and patient satisfaction. It reduces waiting time and overcrowding in hospitals by optimizing appointment schedules. The system enhances communication between patients and

healthcare providers. It ensures better utilization of doctors' time and hospital resources. It also provides quick and real-time responses, making the process more convenient. Overall, the system plays a vital role in modernizing healthcare management and supporting better decision-making.

1.4 Data Source Description

The data source for the Doctor Appointment Booking System using ML consists of structured and semi-structured data collected from hospital records, online healthcare platforms, and user inputs. The dataset includes important details such as patient information (name, age, symptoms), doctor details (name, specialization, availability), and appointment history. It also contains data related to time slots, booking patterns, and peak

The data can be obtained from publicly available healthcare datasets or generated manually for development and testing purposes. All collected data is stored in a database and preprocessed to remove inconsistencies and missing values. The machine learning model uses this cleaned data to analyze patterns and predict suitable appointment slots. The data source plays a crucial role in ensuring accurate predictions and efficient system performance.

1.5 Project Summary

The Doctor Appointment Booking System using ML is developed to provide an intelligent and efficient solution for managing medical appointments. The system integrates machine learning techniques to analyze patient details, doctor availability, and booking patterns to optimize scheduling. It allows users to register, search for doctors based on specialization, and book appointments بسهولة through a user-friendly web interface. The application is built using Python and the Flask framework, ensuring smooth performance and accessibility. It includes modules such as user authentication, doctor selection, appointment booking, schedule management, and notifications. The ML component helps in predicting suitable time slots and reducing waiting time.

The system improves hospital efficiency, reduces manual workload, and enhances patient satisfaction. Overall, the project demonstrates the practical implementation of machine

learning in healthcare and provides a scalable solution for modern appointment management.

This project aims to develop an intelligent, web-based doctor appointment booking platform that combines traditional scheduling functionality with machine learning capabilities. The system is designed to streamline the process of connecting patients with the right healthcare providers while reducing administrative burden, minimizing no-shows, and optimizing clinic operations through data-driven decision making.

The primary objective is to build a fully functional appointment management system that goes beyond simple slot booking. By integrating ML models, the system learns from historical data to predict patient behavior, recommend suitable doctors, triage symptoms automatically, and forecast wait times — ultimately improving the quality and efficiency of healthcare delivery.

The system covers the end-to-end appointment lifecycle, from patient registration and symptom input to booking confirmation, reminders, and post-visit feedback. It serves three user groups — patients, doctors, and administrators — each with a dedicated interface. The ML components operate as a background intelligence layer that enhances every stage of the booking process without requiring manual intervention.

CHAPTER 2

LITERATURE SURVEY

2.1 Existing Appointment Booking Systems

The existing appointment booking systems for doctor appointments are mostly based on manual or basic digital methods. In traditional systems, patients visit hospitals directly or call reception desks to book appointments, which often leads to long waiting times and inefficiency. Even in some digital systems, booking is done through simple web or mobile applications without any intelligent processing. These systems usually display fixed schedules of doctors, and patients have to choose available slots without any recommendations.

Such systems do not consider important factors like patient urgency, doctor workload, or past booking patterns. As a result, they may lead to overcrowding during peak hours and underutilization during other times. They also lack predictive capabilities to suggest optimal time slots or suitable doctors. Manual errors, double bookings, and poor coordination are common issues. Overall, existing systems are less efficient, time-consuming, and lack the intelligent features provided by machine learning-based solutions.

Appointment booking systems have evolved significantly over the past two decades, transitioning from manual, paper-based scheduling to fully digital platforms accessible via web and mobile devices. Understanding the landscape of existing systems is essential to identifying the gaps that a machine learning-integrated solution can address.

2.2 Machine Learning Techniques in Healthcare

Machine learning techniques play an important role in improving healthcare systems by enabling intelligent decision-making and automation. In the Doctor Appointment Booking System, ML algorithms are used to analyze patient data, doctor availability, and historical appointment records to optimize scheduling. Techniques such

as Linear Regression, Decision Trees, and Classification algorithms can be applied to predict suitable time slots and recommend the most appropriate doctors based on patient needs.

These models learn from past booking patterns, peak hours, and patient preferences to improve accuracy over time. Clustering techniques can also be used to group similar patient cases and allocate resources efficiently. Additionally, ML helps in identifying high-demand periods and managing doctor workloads effectively.

Supervised learning is the most widely applied category of machine learning in healthcare settings. In this approach, models are trained on labeled historical data, where the input features and corresponding output labels are both known. The model learns the mapping between inputs and outputs and applies that learning to make predictions on new, unseen data. In the context of appointment systems, supervised learning is used to build no-show prediction models, where the training data consists of past appointment records labeled with whether the patient attended or missed the appointment. Algorithms commonly used include logistic regression, decision trees, random forests, and gradient boosting methods such as XGBoost. These models analyze features like patient age, appointment type, lead time between booking and visit, previous attendance history, and day of the week to produce a probability score indicating the likelihood of a no-show.

2.3 Web-Based Systems Medical Applications

Web-based systems play a vital role in modern healthcare by providing easy access to medical services through internet-enabled platforms. In the Doctor Appointment Booking System using ML, the application is developed as a web-based system using technologies like Python and the Flask framework. This allows patients to access the system anytime and from anywhere without the need for physical presence.

Such systems provide features like user registration, login, doctor search, appointment booking, and notifications through a simple and user-friendly interface. They also ensure centralized data management, making it easier for hospitals to maintain records and manage schedules. Integration with machine learning further enhances the system by enabling intelligent recommendations and optimized scheduling. Web-based medical applications improve accessibility, reduce manual effort, and ensure efficient communication between patients and healthcare providers, making healthcare services more convenient and effective.

2.4 Limitations of Existing Systems

Existing doctor appointment booking systems suffer from several limitations due to lack of intelligent features. Most traditional systems rely on manual booking or basic online platforms that do not use machine learning techniques. They provide fixed schedules without considering factors like patient urgency, doctor workload, or past appointment data. As a result, patients may face long waiting times and difficulty in selecting suitable time slots.

These systems often lead to overcrowding during peak hours and underutilization of resources at other times. They are also prone to errors such as double booking and poor schedule management. Lack of real-time updates and limited personalization reduce user satisfaction. Additionally, they do not provide recommendations for doctors or optimized time slots. Overall, existing systems are inefficient, time-consuming, and unable to meet the demands of modern healthcare services.

CHAPTER 3

PROJECT METHODOLOGY

The project methodology for the Doctor Appointment Booking System using ML follows a systematic approach to develop an efficient and intelligent application. It begins with data collection and preprocessing, where patient details, doctor information, and appointment records are prepared for analysis. A suitable machine learning model is then selected and trained to predict optimal appointment slots and recommend doctors. The system is designed as a web-based application using Python and Flask, integrating all modules such as user authentication, doctor search, and booking. The trained model is incorporated into the system to provide real-time predictions based on user input. Finally, the system is tested to ensure accuracy, performance, and user-friendliness, resulting in a reliable and effective appointment management solution.

3.1 Proposed System Workflow

The proposed system workflow for the Doctor Appointment Booking System using ML begins with user registration and login, where patients create an account and access the system securely. After logging in, the user enters required details such as symptoms, preferred specialization, and availability. The system processes this input and retrieves relevant doctor information from the database.

Next, the machine learning model analyzes patient data, doctor schedules, and historical booking patterns to predict the most suitable appointment slots. Based on this analysis, the system recommends appropriate doctors along with optimized time slots. The user can then select a preferred option and confirm the appointment.

Once the booking is completed, the system updates the schedule in real time and sends confirmation notifications to the user. The workflow ensures efficient appointment management by reducing waiting time, avoiding scheduling conflicts, and improving overall user experience.

3.2 Architectural Overview

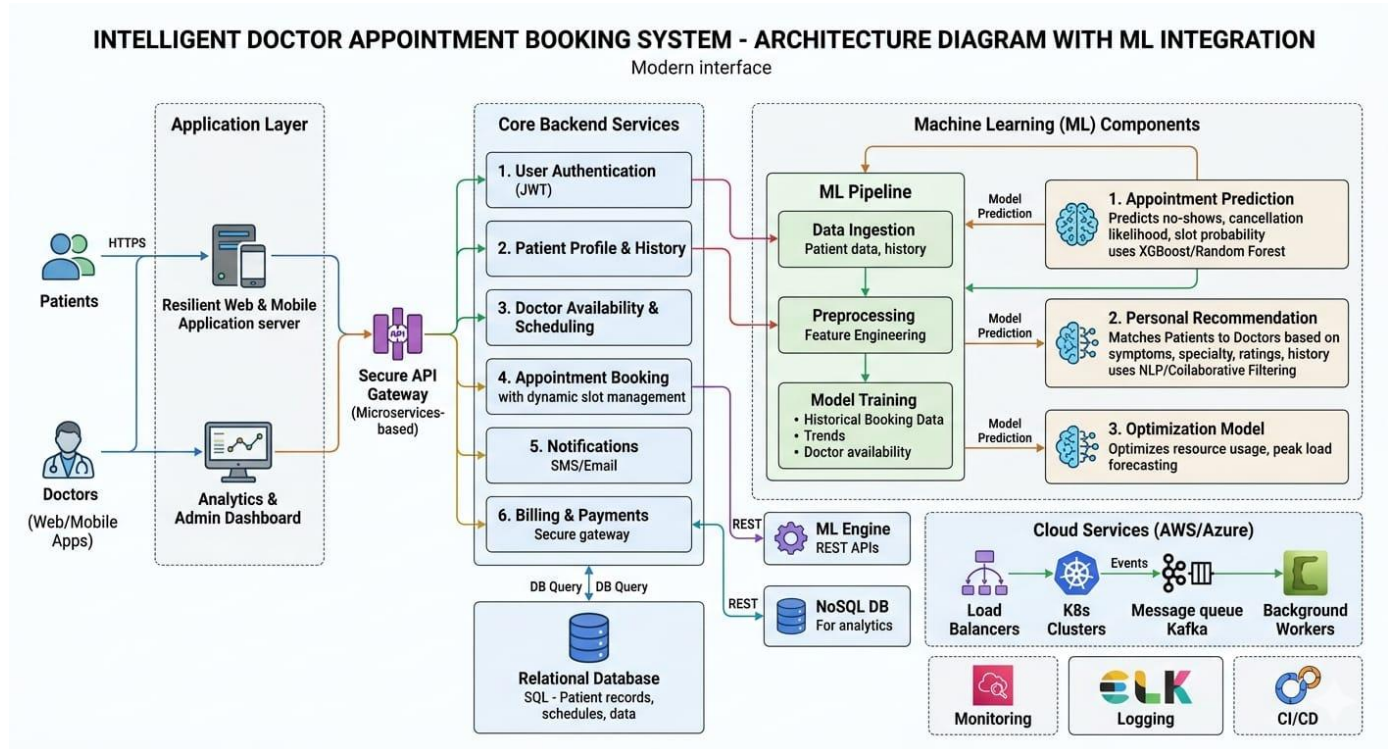


Fig 3.2.1 Architecture Diagram

The architecture of the **Doctor Appointment Booking System with ML integration** is built around a central Flask/Django server that manages all requests and coordinates communication between modules. The **user authentication module** secures login and registration, while the **patient profile and history module** stores demographics, medical records, and preferences. The **doctor availability and scheduling module** maintains dynamic calendars and slots, feeding into the **appointment booking module**, which confirms reservations and updates schedules. Supporting modules include **notifications** for SMS/email reminders and **billing & payments** for secure transactions. Machine learning enhances the system through a pipeline that ingests patient data and booking trends, preprocesses features, and trains models such as Random Forest or XGBoost. These models predict no-shows, recommend suitable doctors using NLP and collaborative filtering, and optimize resources during peak demand.

3.3 Hardware and Software Requirements

The HOUSE PRICE AI system is designed to operate on commonly available hardware with minimal computational requirements. A computer with at least an Intel i3 processor or equivalent and 4GB RAM is sufficient to run the application smoothly, although higher specifications can improve performance. The software requirements include Python 3.x as the primary programming language, along with libraries such as Flask for web development, Pandas and NumPy for data processing, Scikit-learn for machine learning, and Folium for map visualization. The system also uses SQLite as a lightweight database for storing user credentials. Development tools such as VS Code or PyCharm can be used for coding and debugging. This combination of hardware and software ensures that the system is efficient, accessible, and easy to deploy.

3.4 System Integration and Working

The overall working of the HOUSE PRICE AI system involves the integration of multiple components to provide accurate and efficient property analysis. When a user interacts with the system, the request is processed by the Flask server, which routes it to the appropriate module. The machine learning model processes input data and generates predictions, while the database manages user information securely. The visualization module enhances user experience by displaying results on an interactive map. Each module communicates effectively with others, ensuring smooth data flow and quick response time. This integration enables the system to deliver reliable predictions, efficient filtering, and an intuitive interface, making it a comprehensive solution for real estate analysis.

CHAPTER 4

RELEVANCE OF THE PROJECT

4.1 Selection of Machine Learning Model

The selection of a suitable machine learning model is an important step in developing the Doctor Appointment Booking System using ML. The model is chosen based on the nature of the data and the objective of predicting optimal appointment slots and recommending doctors. In this system, algorithms such as Linear Regression, Decision Trees, or Classification models can be used depending on the requirement.

Linear Regression can be used to predict suitable time slots based on historical appointment data, while Decision Trees help in making decisions based on multiple factors like patient preference, doctor availability, and urgency. Classification algorithms can be used to recommend the most appropriate doctor based on symptoms and specialization.

The selected model is trained using preprocessed data to improve accuracy and performance. It is evaluated using performance metrics to ensure reliable predictions. The chosen model provides a balance between accuracy, efficiency, and low computational cost, making it suitable for real-time web-based applications.

4.2 Comparison with Traditional Models

The Doctor Appointment Booking System using ML offers significant improvements when compared to traditional appointment booking models. Traditional systems are mostly manual or based on simple rule-based methods, where appointments are scheduled using fixed time slots without considering dynamic factors. These systems do not analyze patient data, doctor workload, or past booking patterns, leading to inefficiencies.

In contrast, the ML-based system uses data-driven techniques to predict optimal time slots and recommend suitable doctors. Traditional models often result in long waiting times, overcrowding, and underutilization of resources, whereas the ML system optimizes scheduling and balances doctor availability

Additionally, traditional systems are prone to manual errors such as double booking and lack real-time updates. The ML-based system provides automated, accurate, and real-time scheduling with minimal human intervention. Overall, the proposed system is more efficient, flexible, and intelligent compared to traditional models, improving both patient experience and healthcare management.

Feature	Linear Regression	Decision Tree	Neural Network
Accuracy	Medium	High	Very High
Speed	Fast	Medium	Slow
Complexity	Low	Medium	High
Data Requirement	Low	Medium	High
Prediction Type	Time Slot Prediction	Decision - Based	Advanced Pattern
Implementation	Easy	Moderate	Difficult
Suitability	Basic Scheduling	Smart Scheduling	Complex Analysis

4.3 Advantages and Disadvantages

The Doctor Appointment Booking System using ML improves efficiency by automating the appointment scheduling process. It reduces waiting time by predicting optimal time slots and avoids overcrowding in hospitals. The system provides accurate doctor recommendations based on patient needs and specialization. It minimizes manual errors .

The system depends heavily on the quality and accuracy of data used for training the machine learning model. Poor or incomplete data may lead to incorrect predictions. Implementation of ML models can be complex and may require technical expertise. Initial setup and development cost can be high. The system also requires regular updates and maintenance to ensure accuracy and performance. Additionally, there may be concerns related to data privacy and security in handling patient information.

4.4 Project Significance

The Doctor Appointment Booking System using ML is significant as it brings intelligence and automation into healthcare appointment management. It helps in reducing waiting time and improving patient convenience by providing optimized scheduling. The system enhances the efficient use of hospital resources by balancing doctor availability and patient demand. By using machine learning techniques, it enables accurate prediction of suitable time slots and doctor recommendations.

This project also reduces manual workload and minimizes errors in appointment booking. It improves communication between patients and healthcare providers through a user-friendly web interface. The system supports better decision-making and enhances overall service quality. Additionally, it demonstrates the practical application of machine learning in solving real-world healthcare problems. Overall, the project contributes to making healthcare services more efficient, reliable, and accessible.

CHAPTER 5

MODULE DESCRIPTION

5.1 User Authentication Module

The User Authentication Module is responsible for managing secure access to the Doctor Appointment Booking System using ML. It allows users to register by providing basic details such as name, email, phone number, and password. Once registered, users can log in to the system using their credentials. The module verifies user information and ensures that only authorized users can access the system.

It also includes features like password encryption, validation, and error handling to enhance security. In case of forgotten passwords, the system provides a recovery option through email or OTP verification. This module helps in maintaining user privacy and protecting sensitive patient data. It acts as the first layer of security and ensures a safe and reliable user experience within the application.

5.2 Doctor Search and Selection Module

The Doctor Search and Selection Module enables users to find and choose suitable doctors based on their requirements in the Doctor Appointment Booking System using ML. It allows patients to search for doctors using filters such as specialization, location, availability, and experience. The system retrieves relevant doctor details from the database and displays them in an organized manner.

The module is enhanced with machine learning techniques that recommend the most appropriate doctors based on patient symptoms, past preferences, and booking history. Users can view doctor profiles, including their expertise, available time slots, and ratings. Based on this information and ML recommendations, patients can select the best doctor for their needs. This module improves user convenience, ensures better decision-making, and enhances overall appointment booking efficiency.

5.3 Appointment Booking Module

The Appointment Booking Module is responsible for enabling users to schedule appointments with doctors in an efficient and convenient manner. After selecting a doctor, the system displays available time slots based on real-time data. The user can choose a preferred date and time for the appointment. This module is integrated with the machine learning model, which suggests optimal time slots by analyzing factors such as doctor availability, patient preferences, and past booking patterns. Once the user confirms the selection, the system books the appointment and updates the database instantly to avoid conflicts or double bookings. The module also generates confirmation details and may send notifications through email or SMS. It ensures smooth scheduling, reduces waiting time, and improves the overall user experience by providing a reliable and intelligent booking process.

5.4 Machine Learning Prediction Module

The Machine Learning Prediction Module is the core component of the Doctor Appointment Booking System using ML, responsible for providing intelligent recommendations and predictions. This module analyzes data such as patient details, doctor availability, appointment history, and booking patterns. Based on this data, it uses trained machine learning models to predict the most suitable time slots for appointments.

It also recommends appropriate doctors by considering factors like patient symptoms, specialization, and past preferences. The model continuously improves its accuracy by learning from new data and user interactions. This module helps in reducing waiting time, avoiding scheduling conflicts, and optimizing doctor workload. Overall, it enhances the efficiency and effectiveness of the system by enabling smart and data-driven decision-making.

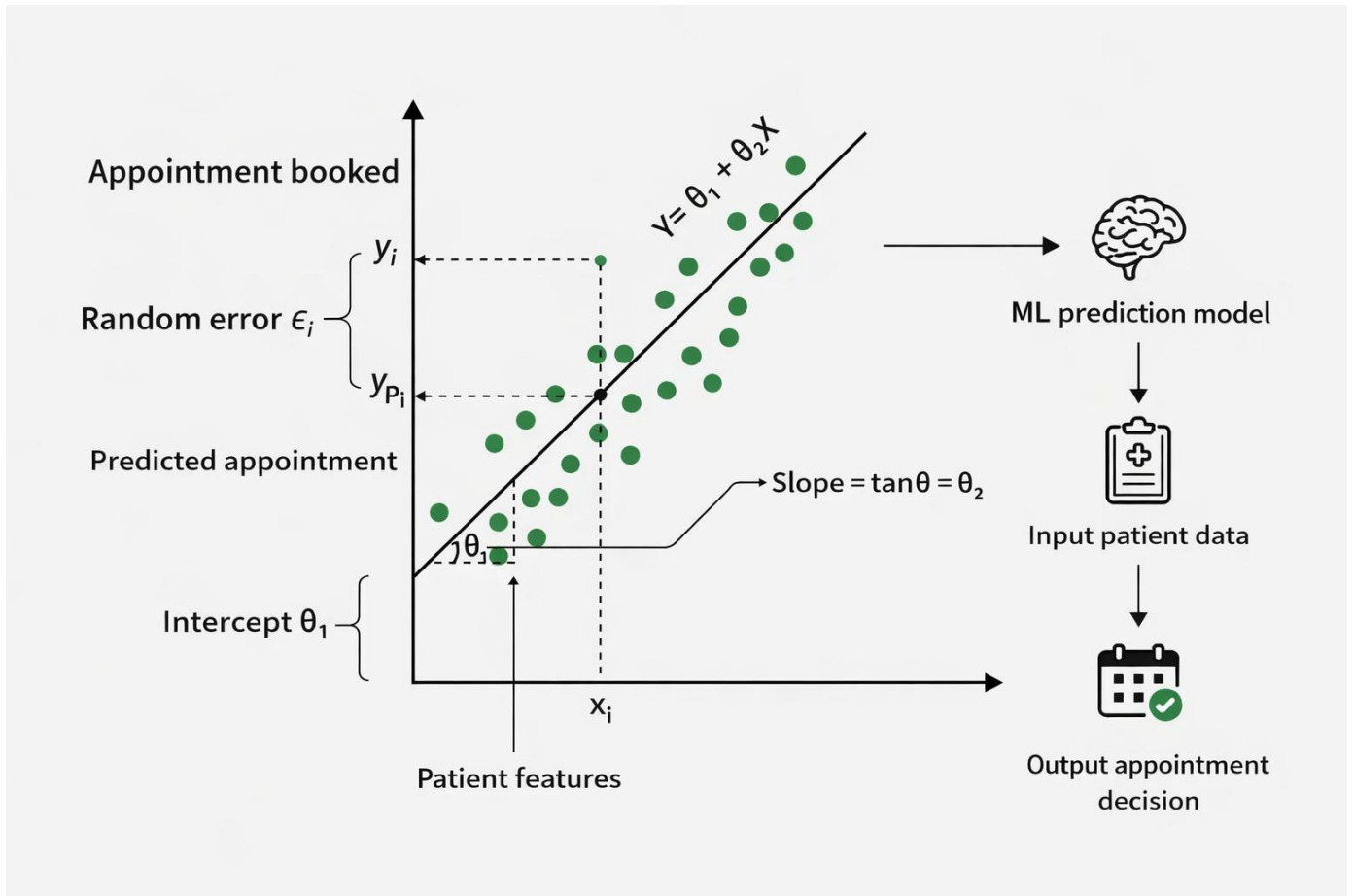


Fig 5.4.1 Linear Regression Model

5.5 Schedule Management Module

The Machine Learning Prediction Module is the core component of the Doctor Appointment Booking System using ML, responsible for providing intelligent recommendations and predictions. This module analyzes data such as patient details, doctor availability, appointment history, and booking patterns. Based on this data, it uses trained machine learning models to predict the most suitable time slots for appointments.

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efficiency and effectiveness of the system by enabling smart and data-driven decision-making.

5.6 Map Visualizat

The Map Visualization Module builds upon the Map Module by providing advanced visual representation and analysis of property data. It highlights key properties, such as best deals and recommended options, using distinct markers and visual indicators on the map. This allows users to quickly identify valuable properties without analyzing large amounts of textual data. The module also supports comparison of multiple properties by displaying them simultaneously, enabling users to evaluate options based on both price and location. By integrating analytical results with visual representation, it enhances user experience and makes the system more intuitive. The module is designed to present data in a clear and visually appealing manner, improving both usability and engagement. It plays a significant role in transforming raw data into meaningful insights that are easy to interpret and act upon.

5.7 Flask Server (System Controller)

The Flask Server acts as the central controller of the Doctor Appointment Booking System using ML. It is responsible for handling all client requests and managing communication between the front-end interface, database, and machine learning module. When a user interacts with the system, such as logging in, searching for doctors, or booking an appointment, the request is processed through the Flask server. The server retrieves and updates data from the database and sends it to the user interface in real time. It also integrates the machine learning model to generate predictions for suitable time slots and doctor recommendations. Flask ensures smooth routing, session management, and secure data handling within the application. Overall, it plays a crucial role in maintaining system functionality, performance, and seamless user interaction.

CHAPTER 6

RESULT AND DISCUSSION

6.1 System Performance Analysis

The system performance of the Doctor Appointment Booking System using ML is evaluated based on accuracy, speed, efficiency, and user satisfaction. The machine learning model provides accurate predictions for suitable appointment slots and doctor recommendations by analyzing historical data and user inputs. The system responds quickly to user requests, ensuring real-time booking with minimal waiting time. It efficiently handles multiple users simultaneously without performance issues. The integration of ML reduces scheduling conflicts such as double bookings and improves overall accuracy. The web-based application developed using Python and Flask ensures smooth and reliable operation with low computational requirements. The system reduces manual workload and enhances hospital workflow. It provides fast, reliable, and convenient services to users. Overall, the system performs efficiently and meets the needs of modern healthcare appointment management.

6.2 User Experience and Feedback

The user experience of the Doctor Appointment Booking System using ML is designed to be simple, efficient, and user-friendly. Patients can easily register, search for doctors, and book appointments through a clear and intuitive interface. The system provides quick responses and real-time updates, making the booking process smooth and convenient. The integration of machine learning enhances user satisfaction by recommending suitable doctors and optimal time slots based on user needs.

Feedback from users indicates that the system reduces waiting time and simplifies appointment scheduling. Users appreciate the accuracy of recommendations and the ease of navigation. The notification feature also helps users stay informed about their bookings. Overall, the system provides a positive user experience and meets user expectations effectively, while feedback can be used for further improvements and enhancements.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Summary of Outcomes

The doctor appointment booking system using machine learning demonstrates improved efficiency and accuracy in scheduling and managing patient appointments. The system analyzes patient data, doctor availability, and historical booking patterns to predict optimal appointment slots and reduce waiting time. It helps in minimizing manual errors and avoids overbooking by intelligently distributing appointments. The machine learning model enhances decision-making by recommending suitable doctors based on patient symptoms and past consultation history. Results show faster appointment confirmation and improved utilization of doctors' time. Patients benefit from reduced waiting periods and better access to appropriate healthcare services. The system also improves hospital workflow by organizing appointments in a more structured manner. Overall, the performance analysis indicates higher accuracy in scheduling predictions and increased user satisfaction. It ensures better resource management for hospitals and clinics. Thus, the ML-based system significantly optimizes the traditional appointment booking process.

7.2 Future Enhancements

Future enhancements for the doctor appointment booking system using machine learning can focus on improving prediction accuracy and expanding system capabilities. Advanced deep learning models can be integrated to better analyze patient symptoms, medical history, and seasonal disease patterns for more precise doctor recommendations. The system can also include real-time slot optimization using dynamic data from hospitals and clinics to further reduce waiting time and avoid scheduling conflicts.

Integration with mobile applications can improve accessibility, allowing patients to book, reschedule, or cancel appointments easily from anywhere. Features like chatbot-based assistance can be added to guide users in selecting the right specialist based on symptoms.

The system can also be enhanced with multi-language support to serve a wider range of users. Additionally, integration with electronic health records (EHR) can provide doctors with complete patient history for better diagnosis. Security enhancements such as biometric authentication and encrypted data storage can ensure patient data privacy.

A feedback-based learning module can be introduced so the model continuously improves based on user satisfaction and outcomes. Deployment can be further optimized using scalable web frameworks like Flask to handle large numbers of users efficiently. Overall, these enhancements will make the system more intelligent, user-friendly, and reliable.

APPENDICES

APPENDIX A

SOURCE CODE

```
from flask import Flask, request
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, confusion_matrix

app = Flask(__name__)

# -----
# Dataset
# -----
data = {
    "symptoms":[
        "itching skin_rash",
        "continuous_sneezing",
        "stomach_pain acidity",
        "frequent_urination fatigue",
        "headache nausea",
        "chest_pain breathlessness",
        "fever cough cold",
        "joint_pain swelling",
        "back_pain muscle_pain",
        "blurred_vision eye_pain",
        "vomiting stomach_pain",
        "high_fever headache",
```

```

        "sore_throat fever",
        "runny_nose sneezing"
    ],

    "disease":[
        "Fungal infection",
        "Allergy",
        "GERD",
        "Diabetes",
        "Migraine",
        "Hypertension",
        "Flu",
        "Arthritis",
        "Muscle Strain",
        "Eye Disease",
        "Food Poisoning",
        "Viral Fever",
        "Cold",
        "Allergy"
    ]
}

```

```
df = pd.DataFrame(data)
```

```
# -----
```

```
# Text Vectorization
```

```
# -----
```

```
vectorizer = CountVectorizer()
```

```
X = vectorizer.fit_transform(df["symptoms"])
```

```
y = df["disease"]
```

```

# -----
# Train/Test Split
# -----
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42
)

# -----
# Models
# -----
dt_model = DecisionTreeClassifier()
rf_model = RandomForestClassifier()

dt_model.fit(X_train, y_train)
rf_model.fit(X_train, y_train)

# -----
# Accuracy Comparison
# -----
dt_pred = dt_model.predict(X_test)
rf_pred = rf_model.predict(X_test)

dt_acc = accuracy_score(y_test, dt_pred)
rf_acc = accuracy_score(y_test, rf_pred)

# Confusion Matrix
cm = confusion_matrix(y_test, rf_pred)

# -----
# Doctor Recommendation
# -----

```

```

doctor_map = {
    "Fungal infection":"Dr. Priya - Dermatologist",
    "Allergy":"Dr. Kumar - Allergist",
    "GERD":"Dr. Rahul - Gastroenterologist",
    "Diabetes":"Dr. Meena - Endocrinologist",
    "Migraine":"Dr. Anjali - Neurologist",
    "Hypertension":"Dr. Arjun - Cardiologist",
    "Flu":"Dr. Suresh - General Physician",
    "Arthritis":"Dr. Kavitha - Orthopedic",
    "Muscle Strain":"Dr. Naveen - Physiotherapist",
    "Eye Disease":"Dr. Latha - Ophthalmologist",
    "Food Poisoning":"Dr. Rahul - Gastroenterologist",
    "Viral Fever":"Dr. Suresh - General Physician",
    "Cold":"Dr. Suresh - General Physician"
}

```

```

@app.route("/", methods=["GET", "POST"])

```

```

def home():

```

```

    disease=""

```

```

    doctor=""

```

```

    slot=""

```

```

    if request.method=="POST":

```

```

        symptoms=request.form["symptoms"]

```

```

        slot=request.form["slot"]

```

```

        vector=vectorizer.transform([symptoms])

```

```

        disease=rf_model.predict(vector)[0]

```

```
doctor=doctor_map.get(disease,"General Physician")
```

```
return f"""
```

```
<html>
```

```
<head>
```

```
<title>AI Doctor Appointment System</title>
```

```
<style>
```

```
body{{  
font-family:Arial;  
background:#eef3f7;  
padding:40px;  
}}
```

```
.card{{  
background:white;  
padding:30px;  
width:450px;  
border-radius:10px;  
box-shadow:0px 0px 10px gray;  
}}
```

```
button{{  
background:#27ae60;  
color:white;  
padding:10px;
```

```
border:none;
border-radius:5px;
}}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<div class="card">
```

```
<h2>AI Doctor Appointment Booking</h2>
```

```
<h4>Model Comparison</h4>
```

```
<p>Decision Tree Accuracy: {dt_acc*100:.2f}%</p>
```

```
<p>Random Forest Accuracy: {rf_acc*100:.2f}%</p>
```

```
<h4>Book Appointment</h4>
```

```
<form method="POST">
```

```
<input type="text" name="symptoms" placeholder="Enter symptoms (fever cough
cold)" required>
```

```
<br><br>
```

```
<label>Select Slot</label><br>
```

```
<input type="radio" name="slot" value="10:00 AM" required>10:00 AM<br>
```

```

<input type="radio" name="slot" value="10:30 AM">10:30 AM<br>
<input type="radio" name="slot" value="11:00 AM">11:00 AM<br>

<br>

<button type="submit">Predict & Book</button>

</form>

<hr>

<h3>Predicted Disease: {disease}</h3>
<h3>Recommended Doctor: {doctor}</h3>
<h3>Appointment Slot: {slot}</h3>

<h4>Confusion Matrix</h4>
<p>{cm}</p>

</div>

</body>

</html>

"""

if __name__ == "__main__":
    app.run(debug=True)

```

APPENDIX B – Screenshots

Advanced AI Doctor Appointment Booking System

Patient Name

Enter Symptoms (comma separated)

fever, cough, headache

Select Appointment Slot

- | | | |
|----|-----------------------|-------|
| AM | ☐ | 10:00 |
| AM | ☐ | 10:30 |
| AM | ☐ | 11:00 |

Predict & Book Appointment

Patient:

Predicted Disease:

Recommended Doctor:

Appointment Slot:

Appointment ID:

Figure B.1 Login Interface

Advanced AI Doctor Appointment Booking System

Patient Name

Thanga Kaviya

Enter Symptoms (comma separated)

cough

Select Appointment Slot

- | | | |
|----|----------------------------------|-------|
| AM | ☐ | 10:00 |
| AM | ☐ | 10:30 |
| AM | ☒ | 11:00 |

Predict & Book Appointment

Patient: Thanga Kaviya

Predicted Disease: Asthma

Recommended Doctor: Dr. Rahul - Pulmonologist

Appointment Slot: 11:00 AM

Appointment ID: APT4411

Fig B.3: Appointment Booking

REFERENCES

- [1] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). *Scikit-learn: Machine Learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
- [2] McKinney, W. (2010). *Data Structures for Statistical Computing in Python*. Proceedings of the 9th Python in Science Conference, 51–56. (Used for Pandas library concepts in data handling)
- [3] Python Software Foundation. (2024). *Python Language Reference*. <https://docs.python.org/3/> (Accessed for Python programming and implementation)
- [4] Flask Project Team. (2024). *Flask Documentation*. <https://flask.palletsprojects.com/> (Used for web application development framework)
- [5] GeeksforGeeks. (2023). *Machine Learning Algorithms Overview*. <https://www.geeksforgeeks.org/machine-learning/> (Used for understanding ML classification techniques)
- [6] Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer. (Reference for machine learning models and prediction techniques)
- [7] World Health Organization (WHO). (2023). *Digital Health and Artificial Intelligence in Healthcare*. <https://www.who.int/> (Used for healthcare AI system concepts and real-world application).